

# Ian Li

Linkedin: <https://linkedin.com/in/ianayl> | Github: <https://github.com/ianayl> | Email: [ianayl.work@gmail.com](mailto:ianayl.work@gmail.com)

## Education

University of Western Ontario - Honours Computer Science – 3.9/4.0 GPA

2021 - Apr. 2026

## Technical Skills

|                              |                                                                                      |
|------------------------------|--------------------------------------------------------------------------------------|
| <b>Programming Languages</b> | C, C++, Python, CUDA, SYCL (DPC++), Bash / POSIX Shell (and more, feel free to ask!) |
| <b>Tooling</b>               | LLVM, MLIR, Linux, Clang (Frontend), Github Actions, GDB/LLDB, Nix/NixOS             |
| <b>Interests</b>             | (GP)GPU compilers, AI/ML compilers, Heterogenous Computing                           |

## Experience

### Compiler Engineer Intern - Intel

18 months – May 2024 - Oct. 2025

Contributing to upstream **MLIR** and Intel MLIR extensions (IMEX) for **SPIRV IR** generation:

- Added new lowerings to **MLIR** ArithToSPIRV pass for converting `arith.index.cast` ops to legal SPIRV ops
- Developing **codegen** for native (hardware) math operations support in IMEX, enabling *fast* math operations on Intel GPUs

Worked on **SYCL runtime (C/C++)** for the DPC++ ([intel/llvm](#)) SYCL compiler, managing construction/scheduling of GPU kernels:

- Implemented **GPU**-host shared memory prefetch to DPC++ for *eliminating* latency from page faults (DPC++, Unified Runtime)
- Debugged critical issues in DPC++ compiler across diverse environments, ranging from deadlocks when running distributed PyTorch models (*Intel Pytorch Extension*, Level Zero) to debugging faulty GPU IR behavior (GenISA, SPIRV, IGC)
- Developed CI for **performance benchmarking** and automatic **regression detection** for [intel/llvm](#) (SYCL, Python, Shell, C++)

### Research Assistant - Ontario Research Centre for Computer Algebra

8 months – Sep. 2023 - Apr. 2024

Researched methods for finding range of **CUDA** kernel launch parameters to optimally **autotune** GPU kernels

- Implemented **LLVM** pass for automatically detecting dimensionality of GPU kernel from **LLVM IR** (via clang), to determine problem space of kernel launch parameters for GPU kernel autotuning
- Researched automatic deduction of *valid* kernel configurations (*constraints*) from GPU kernel source code, in order to speed-up/optimize **GPU kernel autotuning** and avoid erroneous kernel configurations
- Created static code analysis pass (**Clang Frontend**, Quantifier Elimination via [Maple](#)) to automatically derive aforementioned *constraints* from CUDA programs

Goal is a novel GPU autotuning technique requiring *minimal user intervention* while still being *faster* than traditional exhaustive search

### Lead Software Developer - Rentura (startup)

10 months – May 2021 - Feb. 2022

Lead the development and design of a *minimum viable product* (MVP) for a B2C furniture rental platform

- Independently developed server backend (**Javascript**), with a RESTful API using CRUD endpoints secured with *tokens*
- Maintained **MongoDB** instances on a **Linux** server, and designed database schemas for an *order management system*
- Collaboratively developed responsive, mobile-first e-commerce frontend in **React.js** using **Sass** and **Axios** (**MERN** stack)

## Projects

### Experimental ML Compiler - [ianayl/elohssa](#)

May 2024 - Now

- Writing Python bindings (**pybind11**) for ML/tensor operations that can JIT-compile into optimized GPU kernels, akin to PyTorch
- Devising **MLIR** passes for progressively lowering high-level IR from python bindings to GPU dialect to generate GPU binaries

### Experimental Interpreted Language - [ianayl/ex-interpreter](#)

Jun. 2020 - Sep. 2021

- Devised an automatic table-based **lexer generator**, and wrote a **recursive descent** parser following LL grammar
- Wrote an interpreter using a **tree-walk evaluator**, and AST passes to generate CFGs/DAG for IR generation

### Mindless - [ianayl/mindless](#)

Hackwestern 2021

Proof-of-concept *unconditionally secure* biometrics password manager that *generates passwords* on-the-fly instead of storing them

- Devised a method to generate consistent passwords using hashing (SHA256) from *facial landmark data* (using **OpenCV**)
- Developed a password manager (**Python**, **Flask**) implementing said method with a web frontend and API
- Became a finalist project in a 36-hour hackathon (Hackwestern 8) as a one-man team out of 346 teams

### Shell Site Generator - [ianayl/shsg](#)

2020-hiatus

Ultra lightweight and portable static site generator that will run on *anything* with a minimal UNIX-like environment

- Developing a static site generator using **POSIX shell** and **PCRE Regex**: *No dependencies other than a UNIX shell & coreutils!*
- Created a transpiler, transpiling markdown to themed webpages with no installation, compilation, or alternative runtimes required
- Working on removing dependency on PCRE extensions (support for Busybox), and parsing tables and lists in pure regex